# FEDERAL SPECIFICATION

# COMPONENTS FOR INTERIOR ALARM SYSTEM

# BREAKWIRE

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

The complete requirements for procuring the breakwire described herein shall consist of this document and the latest issue of W-A-450/GEN.

## 1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers Type III sensors, breakwire, under Federal Specification W-A-450. The sensors may be affixed to various surfaces.

## 2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

W-A-450/GEN Components for Interior Alarm Systems

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on the date of invitation for bids or request for proposal, shall apply.

Underwriters' Laboratories, Inc.

UL 606 Linings and Screens for Use With Burglar Alarm Systems

UL 681 Installation and Classification of Mercantile and Bank Burglar Alarm Systems

(Application for copies should be addressed to Underwriters' Laboratory, 333 Pfingsten Rd., Northbrook, IL 60062.)

## 3. REQUIREMENTS



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3.1 Materials. Materials for components shall be as specified herein. Material not specified shall be the same as material used commercially and shall be suitable in all respects for the intended purpose.

3.1.1 Wire. Wire shall be hard drawn single strand copper and shall not exceed 0.4 mm diameter. The wire shall have an insulated coating of enamel or varnish.

3.2 Tensile strength. The tensile strength for the breakwire shall not exceed 275 Mpa tensile strength. The break strength shall be 33.6 N and the stress shall be less than 18 N.

3.3 Conductivity. Wire shall be capable of carrying a maximum electrical current of 60 milliamperes at 60 volts with a temperature rise of not more than 1°C.

3.4 Impact strength. The wire shall be capable of withstanding shock twenty times the acceleration due to gravity (196 $m/sec^2$), for a duration of 11 milliseconds and bench handling.

3.5 Operating range. All equipment shall be capable of operating within the following minimum temperature range indicated below :

-28°C to 65.5°C and relative humidity at 95% at 25.5°C.

3.6 Moisture resistance. The breakwire shall be moisture resistant.

3.7 Installation guidelines. Detailed installation procedures for each breakwire application shall be provided by the breakwire manufacturer. The installation guidelines shall be in accordance with UL 606.

3.8 Workmanship. The breakwire shall be manufactured and finished in such a manner as to meet specification requirements.

4. QUALITY ASSURANCE PROVISIONS

4.1 Material inspection. The supplier is responsible for insuring that components and materials used are manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or if none, in accordance with this specification.

4.2 Testing for acceptance. The breakwire shall be tested in accordance with U.L.606 or equal.



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4.3 Testing. Test the breakwire conductivity, tensile strength, and impact strength to ensure the performance parameters identified in sections 3.2, 3.3, and 3.4.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and mailing. The breakwire shall be packaged, packed and marked in accordance with Federal Specification W-A-450.

6. NOTES

6.1 Intended use. Breakwire is an arrangement of electrically conducting wire, screen, tubing placed in front of accessible openings, which initiates an alarm signal when forcing or cutting the lattice breaks the wires. The purpose of breakwire sensors is to provide a means for detecting an intrusion into a normally secure area, when the intrusion is effected by a destructive penetration through a wall, floor, ceiling, door, skylight, or some other similar barrier, on which the grid system is installed. It is configured in fabricating screens, and grids, open wiring, and grooved stripping in various arrays as gridwire necessary to detect surreptitious and forcible entry.

6.2 Ordering data. Purchaser should exercise and desired options permitted herein and include the following information in procurement documents:

(a) Title, symbol, and date of this specification,
(b) Quantity required.
(c) Destination inspection required.
(d) Levels of packaging and packing required.
(e) Whether special marking required.

Preparing activity:
GSA-FSS



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